untitled

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```
import heapq

def a_star(graph, start, goal, h):
    """graph: {node: [(neighbour, cost), ...]},  h: {node: heuristic value}"""
    open_heap = [(h[start], 0, start)]       # (f, g, node)
    g_cost = {start: 0}
    parent = {start: None}
    closed = set()
    while open_heap:
        f, g, node = heapq.heappop(open_heap)
        if node in closed:
            continue
        if node == goal:
            path = []
            while node is not None:
                path.append(node)
                node = parent[node]
            return path[::-1], g
        closed.add(node)
        for nb, cost in graph[node]:
            ng = g + cost
            if ng < g_cost.get(nb, float("inf")):
                g_cost[nb] = ng
                parent[nb] = node
                heapq.heappush(open_heap, (ng + h[nb], ng, nb))
    return None, float("inf")

def a_star_grid(grid, start, goal):
    """grid: list of strings, '#' is a wall. 4-direction moves, cost 1 each."""
    rows, cols = len(grid), len(grid[0])
    h = lambda p: abs(p[0] - goal[0]) + abs(p[1] - goal[1])
    open_heap = [(h(start), 0, start)]
    g_cost = {start: 0}
    parent = {start: None}
    closed = set()
    while open_heap:
        f, g, cur = heapq.heappop(open_heap)
        if cur in closed:
            continue
        if cur == goal:
            path = []
            while cur is not None:
                path.append(cur)
                cur = parent[cur]
            return path[::-1]
        closed.add(cur)
        for dr, dc in ((-1, 0), (1, 0), (0, -1), (0, 1)):
```

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```
Graph demo: path = S -> A -> B -> C -> D -> G | total cost = 10

Grid demo: shortest path length = 15 steps
S***#...
.##*#.#.
.#***#.#.
.#*##.#.
.#*****#G
.#######*
........
```